

Scalable Relational Database System

Design & Implementation Report | Digital Payments Platform

Project Domain	FinTech / Digital Payments
Scope	Database Architecture, OLTP Design & Analytics
Tech Stack	MySQL, SQL, Data Modelling (3NF)
Classification	Internal Technical Report

1. Introduction

This report presents the design and implementation of a scalable relational database system built to power a modern digital payments platform. The architecture is engineered to handle high-volume transaction processing, enforce end-to-end data integrity, and deliver the analytical capabilities needed for confident, data-driven business decisions.

Three core goals shape every design choice: operational efficiency, proactive fraud risk reduction, and an elevated customer experience through reliable, well-structured data.

2. Data Modelling & Architecture

The schema is built on Third Normal Form (3NF) principles to eliminate redundancy and guarantee consistency across all entities. Foreign key constraints enforce referential integrity throughout, ensuring that no orphaned records can compromise data quality.

2.1 Design Philosophy

The Transactions table serves as the central fact table, capturing every financial event. Surrounding dimension-like tables (Users, Merchants) supply the rich contextual data needed for both transactional processing (OLTP) and analytical querying.

2.2 Core Tables

Table	Role	Description
Transactions	Fact *	Central store of all financial activity and lifecycle states
Users	Dimension	User profiles: name, phone, email, KYC status
Wallets	Dimension	Wallet balances and UPI handles linked to users
BankAccounts	Dimension	Bank account details associated with user profiles

Merchants	Dimension	Merchant profiles for payment acceptance flows
TransactionHistory	Audit	Full lifecycle audit trail of transaction state changes
Services	Reference	Offers, coupons, cashback schemes, and loan products
SupportTickets	Operational	User complaints, issue types, and resolution tracking
Feedback	Operational	User ratings and reviews for service quality analysis

3. System Features & Functional Capabilities

3.1 Transaction Processing

The system supports a full range of transaction types — peer-to-peer transfers, bill payments, and cashback processing. Each transaction moves through clearly defined lifecycle states (Processing, Success, Failed), enabling real-time status tracking and operational monitoring at every stage.

3.2 Fraud Detection

A rule-based fraud detection module evaluates every transaction against multiple risk parameters:

- Transaction amount and velocity thresholds
- Device fingerprinting and location signals
- Risk scoring with automated actions: Flag, Block, or Verify

This layer operates in real time, allowing the platform to intercept suspicious activity before it impacts users or revenue.

3.3 Customer Support & Feedback

A unified support framework captures issue types, resolution status, and user ratings in structured tables. This enables systematic service quality analysis and surfaces actionable insights for product and operations teams.

4. Analytical Insights & Business Intelligence

The schema is purpose-built for SQL-driven analytics. Key business intelligence queries include:

Insight Area	What It Measures
Transaction Success Rate	System reliability and payment completion performance
Revenue Computation	Fee aggregation across transaction types and time periods
Fraud Trend Analysis	Risk score distributions and rule trigger frequencies
Customer Satisfaction	Ratings, feedback sentiment, and resolution quality

Together, these metrics support data-driven decision-making and help engineering and product teams identify bottlenecks in operational workflows before they escalate.

5. Operational Efficiency & Performance

Schema design is the first line of defence against poor query performance. Normalised relationships and minimal redundancy keep the data footprint lean. Generated columns (e.g. `net_amount`) pre-compute frequently needed values, eliminating repeated runtime calculations.

Additional performance strategies include:

- Selective indexing on high-frequency query columns: `transaction_status`, `user_id`, and timestamps
- Table partitioning on the Transactions fact table to improve scalability at high volumes
- Query plan optimisation aligned to OLTP access patterns

6. Challenges & Engineering Solutions

Challenge	Solution
Data consistency across related entities	Foreign key constraints and 3NF-normalised schema design
Tracking transaction state changes	TransactionHistory table providing a complete, immutable audit trail
Computing derived metrics efficiently	Generated (computed) columns for net transaction value, reducing runtime overhead

7. Future Enhancements

The following roadmap items are identified for the next phase of development:

- Machine learning integration for predictive, adaptive fraud detection
- Real-time streaming pipelines for sub-second event processing
- Advanced analytics dashboards via Power BI or Tableau
- Multi-currency support for international payment corridors
- Microservices-based architecture for independent service scaling
- REST API integration layer for third-party service connectivity

8. Technology Stack

Component	Technology / Approach
Primary Database	MySQL (Relational)

Query & Analytics	SQL
Schema Design	Data Modelling (3NF Normalisation)
System Design Pattern	OLTP (Online Transaction Processing)

9. Conclusion

This project demonstrates a comprehensive, production-minded approach to designing a fintech-grade database system. By combining rigorous data modelling, performance-aware schema design, and built-in analytical capability, the platform is positioned to scale reliably as transaction volumes grow.

The architecture balances immediate operational needs with a clear path toward advanced capabilities — from machine learning-driven fraud prevention to real-time analytics — making it a strong foundation for the next generation of digital payment infrastructure.

— End of Report —